CompanhHub: Hub Order Demand Forecasting

CSE558 Data Science • Midterm Project Report • Monsoon 2026

Group name DSC 10
Dataset Kaggle hub-order demand dataset supplied for the project
Code `https://github.com/kushiiitd05/CompanhHub-hub-demand-forecasting`

#	Name	Roll no.	Institute email	Contribution
1	Manya Chaturvedi	2023312	manya23312@iiitd.ac.in	
2	Kush Tokas	2023296	Kush23296@iiitd.ac.in	
3	Akash Adur	2024053	akash24053@iiitd.ac.in	
4	Anusha Rana	2023114	anusha23114@iiitd.ac.in	
5	Sanjeev	2023483	sanjeev23483@iiitd.ac.in	

Table 1: Dataset size and encoded scale

Stage	Rows	Columns	Entries
Raw training	970,379	9	8,733,411
After join	970,379	21	20,377,959
Encoded	970,379	35	33,963,265

1. Dataset

The project studies daily order demand at restaurant-service hubs. The training file, `orders_train.csv`, contains 970,379 observed hub–date records and nine columns. The hub metadata file, `hub_metadata.csv`, contains 1,115 hubs and ten metadata columns. The supplied test file, `orders_test.csv`, contains 46,830 rows and eight columns, but it does not contain the response `OrderVolume`; therefore, it is not used for the outcome-based hypothesis tests in this report.

The training period runs from 1 January 2013 to 19 June 2015, spanning 900 distinct dates. There are 1,115 distinct hubs. A row represents an observed hub on a date, rather than every possible hub–date combination. The response variable is `OrderVolume`, the recorded number of orders for a hub on a date.

The encoded representation has $970{,}379 \times 35 = 33{,}963{,}265$ row–attribute entries, exceeding the required three million. Relevant variables include `OrderVolume` (daily demand), `HubID` (hub identifier), date and weekday, promotion status, hub format, competitor distance, competitor opening month/year, and hub loyalty timing fields. Promotion and weekday allow within-hub comparisons; hub format and competitor metadata support cross-hub comparisons.

2. Data preparation

We checked key integrity, duplicate records, missingness, operating status, weekday coding, and the effects of joining and encoding. The key audit found no duplicate `HubID-Date` records, unique hub metadata keys, and no training records whose hub failed to match the metadata.

There are 1,003,500 possible hub–date combinations over the observed date range, compared with 970,379 observed records: 33,121 combinations are absent. These absent combinations are treated as unobserved, not as zero-demand days. Among observed records, 166,269 are marked closed and 804,110 open. The weekday coding was verified as Monday = 1 through Sunday = 7.

Table 2: Preprocessing audit

Issue found	What was done	Why
Hub metadata	Joined by hub key; all training hubs matched	Adds hub context without losing training rows.
Missing metadata	Retained missing values; no blanket imputation	Unknown competitor/loyalty values should not be invented.
Duplicate keys	Checked `HubID-Date`; none found	Prevents repeated hub-days from inflating demand.
Hub-day gaps	Did not fill absent combinations with zero	An absent record does not prove a hub was closed or had zero orders.
Closed/open status	Checked consistency with order volume	No closed row had nonzero orders; 54 open rows had zero orders.
Encoding	Expanded joined data from 21 to 35 columns	Makes categorical variables usable in the encoded representation.

Table 3: Before-and-after size and missingness

Measure	Raw	Joined	Encoded
Rows	970,379	970,379	970,379
Columns	9	21	35
Rows × columns	8,733,411	20,377,959	33,963,265
Missing cells	0	1,589,842	1,589,842

3. What changed after preparation

The join preserved the row count and added 12 columns; encoding added another 14 columns. The missing-cell count increased after joining and did not change during encoding. In the hub metadata, `CompetitorDistance` is missing for three hubs, competitor opening month/year for 354 hubs, and loyalty timing fields for 544 hubs each. We do not describe the resulting data as fully cleaned: missing metadata remain explicit and no arbitrary values are substituted.

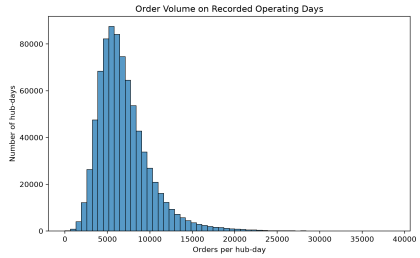


Figure 1: Distribution of daily order volume. Demand is right-skewed, with a long upper tail.

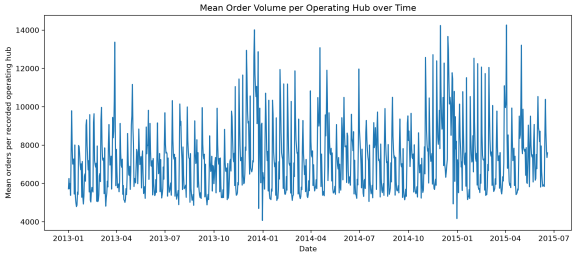


Figure 2: Daily order-volume trend over the training period. The series varies over time; temporal structure may affect comparisons.

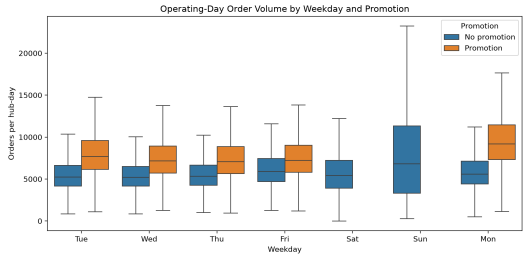


Figure 3: Mean order volume by weekday and promotion status. Promotion-associated differences are positive on weekdays shown.

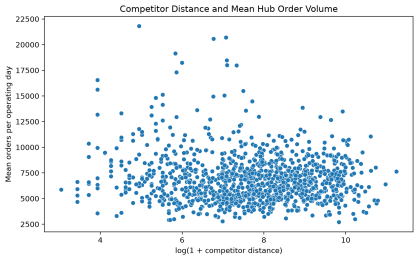


Figure 4: Competitor distance and hub-level mean demand. The rank association is weak and not statistically significant.

4. Basic questions and problem statements

Exploratory analysis shows substantial variation in daily demand. On observed operating days, mean order volume is 6,954.36, median is 6,368, standard deviation is 3,106.67, and the range is 0–38,722. The distribution is right-skewed (skewness 1.593), so the mean is sensitive to high-demand days.

Basic questions include whether promotion-associated demand differences vary by weekday, whether longer gaps in observed hub–date records are associated with hub format, whether competitor distance is associated with average demand, and whether demand changes around a competitor's recorded opening. These questions use different units of analysis (hub-days or hubs) and should not be interpreted as causal effects.

P1 (primary). Is the promotion-associated difference in order volume larger on Mondays than on Fridays, when comparing hubs with observations in both promotion states on both weekdays?

P2. Is the indicator for a long gap in observed hub–date records associated with hub format?

P3. Is competitor distance monotonically associated with a hub's mean order volume?

P4. Does a hub's mean order volume differ between windows before and after a competitor's recorded opening date?

5. Statistical inferences

We use a significance level of $\alpha = 0.05$. Reported p-values are interpreted alongside effect estimates and 95% confidence intervals. Holm adjustment is used across the four primary tests. These are observational analyses; statistical significance does not establish causality.

5.1 Test 1 (P1): Monday versus Friday promotion lift

Hypotheses. Let $\Delta_{h,d}$ be the difference between a hub's mean order volume on promotion and non-promotion observations for weekday d. The paired hub-level contrast is $D_h = \Delta_{h,\text{Mon}} - \Delta_{h,\text{Fri}}$. The null is $H_0 : \mathbb{E}[D_h] = 0$; the two-sided alternative is $H_1 : \mathbb{E}[D_h] \neq 0$. We use a paired t-test across 1,115 hubs with all four groups represented.

Result. The mean paired contrast is +2,168.77 orders (95% CI [2,103.14, 2,234.39]), $t = 64.84$, $p < 0.001$; Holm-adjusted $p < 0.001$. The standardized paired effect size is 1.94. The positive contrast means that the promotion-associated difference is larger on Mondays than Fridays in this sample.

Alternative and caveat. A hub fixed-effects model was fitted as a sensitivity analysis, with standard errors clustered by hub. Its Monday-by-promotion interaction was +2,030.74 orders (95% CI [1,962.95, 2,098.54], $p < 0.001$). Date fixed effects cannot separately identify promotion status because all hubs share the same promotion status on each date in the filtered data. The hub fixed-effects model therefore does not control for all date-specific demand shocks. Promotion may coincide with other calendar effects, so neither estimate is a causal promotion effect.

Table 4: Long-gap indicator by hub format

Hub format	No long gap	Long gap
1	518	84
2	16	1
3	134	14
4	267	81

5.2 Test 2 (P2): long observation gaps and hub format

A long gap is defined by the project's derived indicator in the analysis script. The unit of analysis is one hub ($n = 1{,}115$). We test independence between this binary indicator and the four-level hub-format variable. The null is independence; the alternative is association. We use a Pearson chi-square statistic and a Monte Carlo permutation p-value (20,000 permutations), because at least one expected contingency-table cell is below five.

The observed statistic is $\chi^2 = 21.42$ with 3 degrees of freedom; Cramér's $V = 0.139$. The permutation p-value is approximately 0.00020 and the Holm-adjusted value is approximately 0.00040. This is evidence of an association between hub format and the long-gap indicator, not evidence that format causes missing observations. The smallest expected cell is 2.74, which is why the permutation result is emphasized over the asymptotic approximation.

5.3 Test 3 (P3): competitor distance and mean demand

We aggregate to one observation per hub and use Spearman's rank correlation, which tests for a monotonic association without assuming linearity or normally distributed variables. The hypotheses are $H_0 : \rho_s = 0$ and $H_1 : \rho_s \neq 0$. Competitor distance is available for 1,112 hubs.

The estimated correlation is $\rho_s = -0.0317$ (95% bootstrap CI $[-0.0909,\ 0.0296]$), with $p = 0.291$ and Holm-adjusted $p = 0.291$. The confidence interval includes zero and the estimate is close to zero; this analysis does not provide evidence of a monotonic association. Kendall's $\tau = -0.0193$ ($p = 0.336$) gives the same qualitative conclusion. The relationship is descriptive and may be confounded by hub location, format, and other differences.

5.4 Test 4 (P4): demand around competitor opening

We compare each eligible hub's mean order volume in a pre-opening window with its mean in a post-opening window, using a paired t-test. The opening date is approximated as the 15th day of the month recorded in metadata. For the primary specification, the pre-window spans 105 to 15 days before that date and the post-window spans 15 to 105 days after it. Hubs must have at least 30 operating-day observations in each window, and both windows must fall within the training period. The resulting paired sample contains 120 hubs.

Table 5: Sensitivity analysis for competitor-opening windows

Outer window	Hubs	Mean diff.	95% CI	Wilcoxon p
75 days	138	−433.12	[−603.89, −262.36]	2.92e−6
105 days	120	−482.42	[−650.21, −314.64]	3.37e−7
135 days	112	−476.97	[−639.83, −314.10]	4.95e−7

The mean post-minus-pre difference is −482.42 orders (95% CI $[-650.21,\ -314.64]$), $t = -5.69$, $p = 9.13 \times 10^{-8}$; Holm-adjusted $p = 2.74 \times 10^{-7}$. The observed mean is lower in the post-opening window. Since opening dates are approximate and the comparison lacks a contemporaneous control group, the result should not be interpreted as a causal competitor effect.

The negative difference persists under 60-, 90-, and 120-day pre/post spans, and the paired Wilcoxon tests are also significant. This sensitivity supports robustness to the selected window width, but it does not remove confounding or uncertainty in the approximate opening date.

6. Limitations

- **Observational data:** promotion, hub format, competitor proximity, and opening timing are not randomized. Unmeasured calendar, geographic, operational, or market factors may explain some associations.
- **Incomplete hub–date panel:** 33,121 of 1,003,500 possible hub–date combinations are absent. They are not assumed to be closures or zero-demand days, but this means the observed records may be selective.
- **Metadata missingness:** competitor and loyalty fields have missing values. Tests using competitor distance exclude hubs without a usable distance.
- **Approximate event timing:** competitor opening dates are represented by the 15th of the reported month, and the event-window analysis has no control group.
- **Multiplicity and dependence:** Holm correction is applied across the four main tests, but repeated hub observations and shared calendar conditions still require careful interpretation. Clustered standard errors are used for the fixed-effects sensitivity model.
- **Limited scope:** `orders_test.csv` has no `OrderVolume` outcome, so it is not used for these outcome-based tests. Results describe the available training period and may not generalize to later periods.

Reproducibility

Scripts, data-audit outputs, hypothesis-test results, sensitivity analyses, figures, and instructions are maintained in the project repository: `https://github.com/kushiiitd05/CompanhHub-hub-demand-forecasting`. The analysis

uses the training records and hub metadata described above. All reported numerical results should be regenerated from the scripts in the repository before final submission.